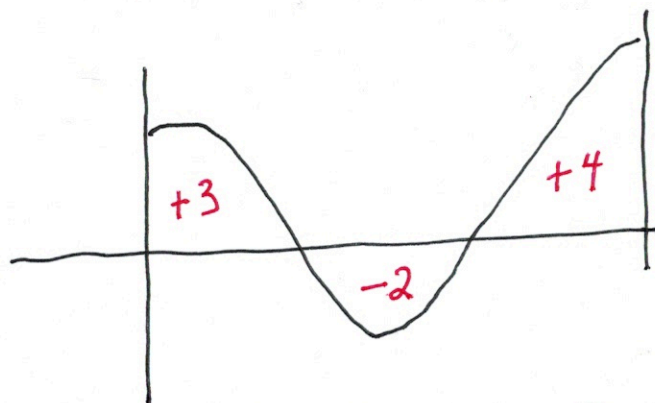


Section 5.4 Indefinite vs. Definite Integrals

$$\int x^2 dx = \frac{x^3}{3} + C$$

$$\int_1^2 x^2 dx = \left. \frac{x^3}{3} \right|_1^2 = \frac{2^3}{3} - \frac{1^3}{3} = \boxed{\frac{7}{3}}$$



$$\text{Net change} = 3 - 2 + 4 = \boxed{5}$$

Ex ① $\int \sec x \cdot \tan x \, dx = \boxed{\sec x + c}$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

② $\int \sin^2 x \, dx = \frac{1}{2} \int (1 - \cos 2x) \, dx$

$$\frac{1}{2} \cdot \left(x - \frac{\sin 2x}{2} \right) + C$$

$$= \frac{1}{2} x - \frac{\sin 2x}{4} + C$$

③ $\int (5x+6)(2x-3) \, dx$

$$\int (10x^2 - 3x - 18) \, dx$$

$$\frac{10x^3}{3} - \frac{3x^2}{2} - 18x + C$$

Ex (4) $\int_1^4 \frac{x-1}{\sqrt{x}} dx$

$$\int_1^4 \frac{x^1}{\sqrt{x}} - \frac{1}{\sqrt{x}} dx$$

$$\int_1^4 \frac{x^{\frac{1}{2}+1}}{\frac{3}{2}} - \frac{x^{-\frac{1}{2}+1}}{\frac{1}{2}} dx$$

$$\left. \frac{2}{3} x^{3/2} - 2 x^{1/2} \right|_1^4$$

$$\frac{2}{3} (4)^{3/2} - 2(4)^{1/2} - \left[\frac{2}{3} - 2 \right]$$

$$\frac{16}{3} - 4 - \frac{2}{3} + 2$$

$$\frac{14}{3} - 2\frac{2}{3}$$

$$\boxed{\frac{8}{3}}$$

Ex (5) Velocity of A baseball
thrown into the air is

$f = \text{dist}$

$f' = \text{velocity}$

$$V = -32t + 50$$

$$\int_0^1 \left(-32 \frac{t^2}{2} + 50\right) dt$$

$$= -16t^2 + 50t \Big|_0^1$$

$$= -16 + 50$$

$$= 34 \text{ feet travelled over first second.}$$

(displacement.)

Ex (6) $w(t) = 5t^2 + 1$ is the

rate water is leaking
out of a storage tank
in gallons per hour.

Find $\int$ interpret

$$\int_0^2 (5t^2 + 1) dt$$

$$= \left. \frac{5t^3}{3} + t \right|_0^2$$

$$= \frac{5(2)^3}{3} + 2$$

$$= \frac{40}{3} + 2 = 15\frac{1}{3} \text{ gallons leaked out over first 2 hours.}$$